Disaster Recovery & Business Continuity

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Background

- Information Systems are vulnerable to a variety of **disruptions**.
  - Mild: Temporary power outages, disk failures etc.
  - Severe: Equipment destruction, fire, natural disasters etc.

- Organizations must have the **ability to withstand hazards** and achieve business objectives through both gradual & sudden changes.

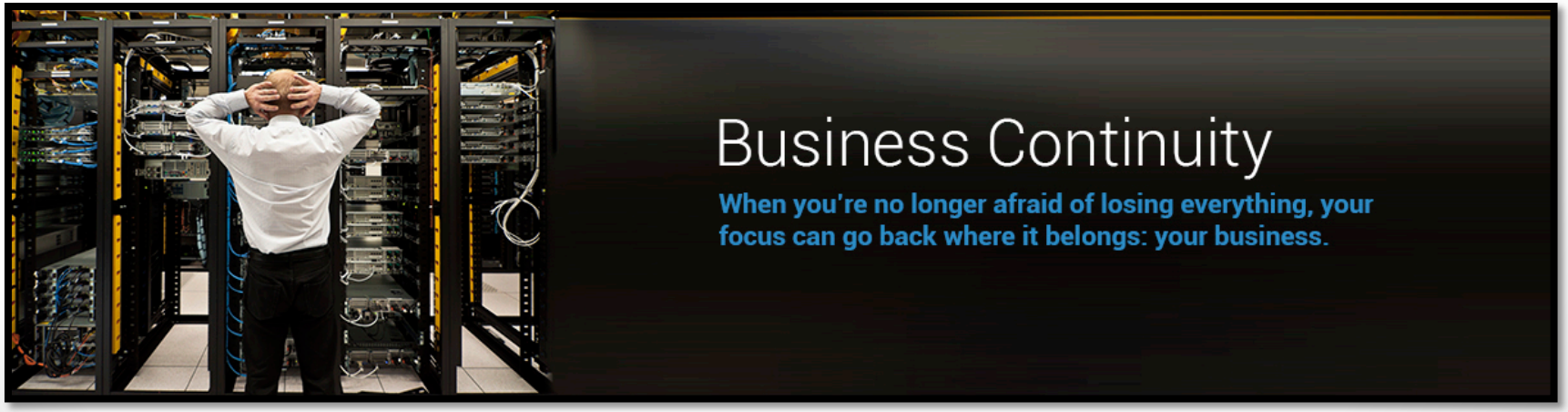
- Focus is on **'Availability'** component of the famous C.I.A

- We achieve this through **'Disaster Recovery Planning'** & **'Business Continuity Planning'**.

Background


FACT:
1 in 4 businesses never re-open their doors after a disaster


90% of businesses fail within 2 years after being struck by a disaster.


Business Continuity
When you're no longer afraid of losing everything, your focus can go back where it belongs: your business.

Background



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NETWORKWORLD Most read:

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UPDATE: Amazon.com suffers outage: Nearly $5M down the drain?

Amazon.com, the major online retailer, is down for 49 minutes

By Brandon Butler | Follow
Network World | Jan 31, 2013 4:55 PM PT

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Definitions

- **Disaster Recovery Planning (DR)**
  - The process of rebuilding your operations or infrastructure after the disaster has passed.

- **Business Continuity Planning (BC)**
  - The activities required to keep your organization running during a period of displacement or interruption of normal operations.

NIST SP 800-34 provides guidelines for implementing DR & BC Strategies!

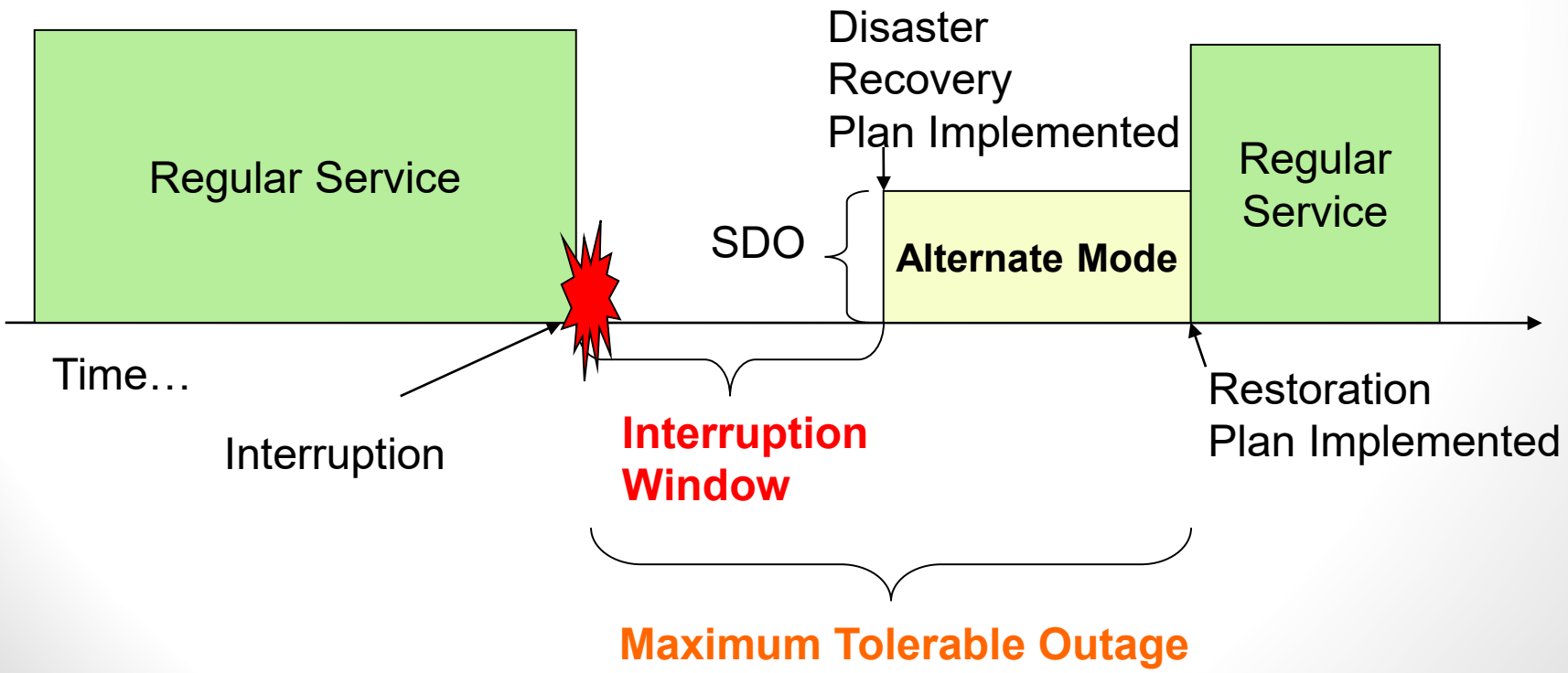
More than half of small to medium-sized enterprises affected by 9/11 did not trade again!

Recovery Time: Terms

Interruption Window: Time duration organization can wait between point of failure and service resumption

Service Delivery Objective (SDO): Level of service in Alternate Mode

Maximum Tolerable Outage: Max time in Alternate Mode



Business Continuity: Why?

- Advancement of IT means businesses nowadays depend heavily on information systems.
- Many businesses **cannot** survive without 24 x 7 operations of IS. (e.g. e-commerce)
- Therefore **traditional disaster recovery** plans which focus on restoring **centralized data & operations** center might not be sufficient.
- More comprehensive and robust **Business Continuity Plan** is needed for critical IS.

Business Continuity : When?

- **Business continuity plan should exist in the event of following disruptions or disasters.**
  - Equipment Failure
  - Disruption of power supply or telecommunication
  - Application failure or database corruption
  - Human error, Sabotage, Vandalism & Strikes
  - Malicious Software (Viruses, Worms, Trojan Horses) Attack
  - Hacking or any other internet attack
  - Social Unrest or Terrorism
  - Fire
  - Natural Disasters (Flood, Earthquake, Hurricanes etc.)

Business Continuity Planning : Team

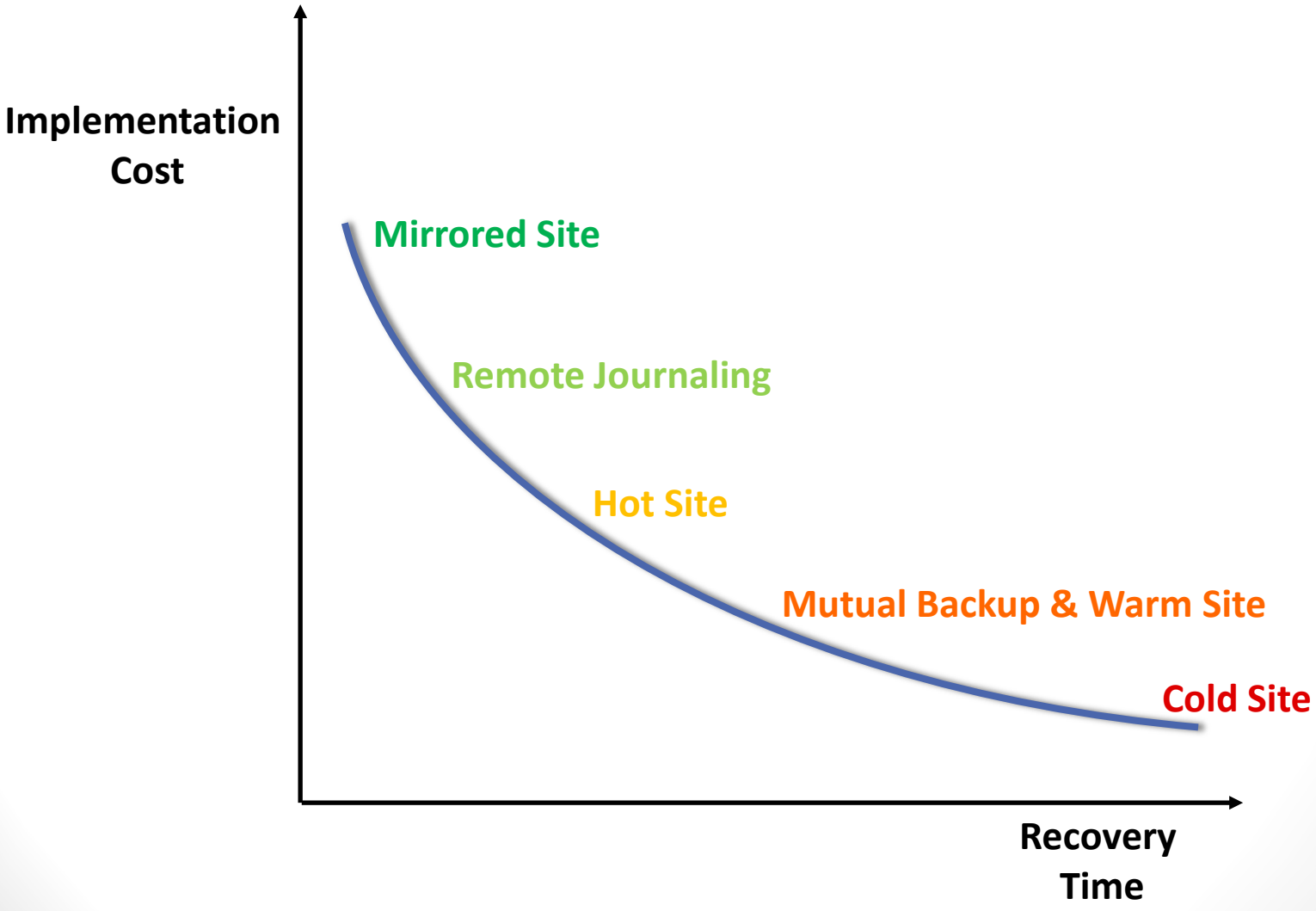
- Information systems have shifted from traditional centralized architecture to distributed and client/server architecture.
- IT department alone **cannot** achieve BCP success
- **All executives, managers, employees must participate**
- **BC/DR Coordinator** is responsible for maintaining the BCP
- He or She will carry out periodical reviews and redistribute document parts to relevant parties

Business Continuity Plan

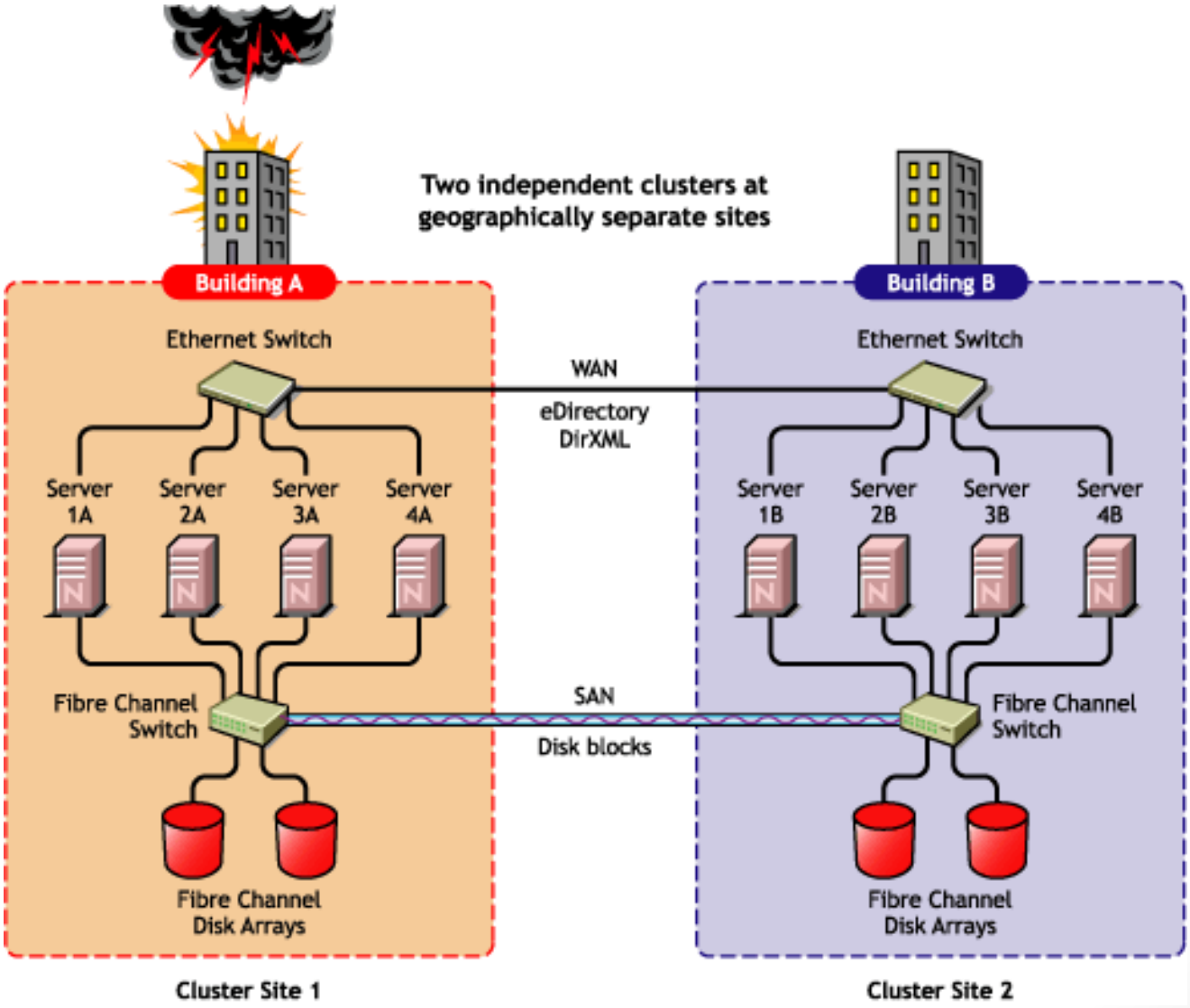
There are different methods an organization can achieve BC

- **Cold Site:** An empty facility located offsite with required infrastructure ready for installation in the event of a disaster.
- **Mutual Backup:** Two organizations with similar system configuration agreeing to serve as backup site to each other.
- **Hot Site:** A site with hardware, software & network installed and compatible to original site.
- **Remote Journaling:** Online transmission of data to backup systems periodically (every few hours) Minimizes loss of data and reduces recovery time.
- **Mirrored Site:** A site equipped with identical facilities to the original site with system mirroring capability. Data is mirrored & backed up immediately. Transparent Recovery.

Business Continuity Plan



Mirrored Site

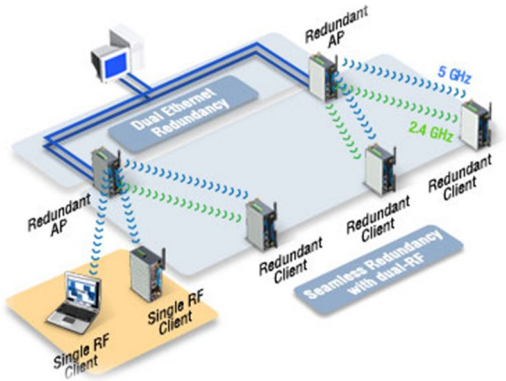


High Availability Solutions

- **RAID:** Local disk redundancy
- **Fault-Tolerant Server**: When primary server fails, backup server resumes service.
- **Distributed Processing**: Distributes load over multiple servers. If server fails, remaining server(s) attempt to carry the full load.
- **Storage Area Network (SAN):** disk network supports remote backups, data sharing and data migration between different geographical locations

Business Continuity Plan

Redundancy Vs. Cost



Vs.



- Right balance between BCP & Cost can be achieved.
- How? (Hint: You already know the answer)

DR & BC Providers

- Organizations can decide to use a facility delivered by a third party BC provider.

- However following areas of concerns should be considered.
  - Floor Space
  - Redundant Equipment
  - Redundant Network Capacity
  - Relationship with vendors to provide replacements or assistance
  - Budgetary Constraints
  - Skilled personnel availability

Click Here: Local DR & BC Provider (LankaCom)

Preparing the BC Plan: Phases

1. **Project Initiation**

   BC objectives are defined and the scope is identified. A committee will be appointed to draw up BC policies.

2. **Business/Risk Analysis**

   Performing the 'Risk Analysis', Considering alternative BC strategies, Cost-benefit analysis, strategy selection & establish the budget.

3. **Design & Development (Plan)**

   BC team is identified and responsibilities are assigned. Develop BC strategy and action plan and plan activation criteria.

4. **Implementation (Plan)**

   Prepare disaster response & recovery procedures. Vendor contracts prepared and recovery resources are purchased. Ensure that recovery team on alert.

5. **Testing** - Exercise scenarios periodically & produce BC reports & evaluate.

6. **Maintenance** - Reviewing & constantly updating/improving the BC plan.

Concerns for a BCP/DR Plan

- Evacuation plan: People's lives always take **first priority**
- Disaster declaration: Who, how, for what?
- Responsibility: Who covers necessary disaster recovery functions
- Procedures for Disaster Recovery
- Procedures for Alternate Mode operation
  - Resource Allocation: During recovery & continued operation

*Copies of the plan should be off-site

Legally Obligated

- In some organizations business analysis [2] is **not** the only factor that determines BC Strategy.
- They are **legally obligated** by regulators to provide certain levels of protection to client data.
- Organizations who have direct **public interest** (such as banks) have legal obligations to implement DR & BC strategies.
- http://www.slcert.gov.lk/ Provides consultancy on DR & BC planning.

Click Here: Central Bank Guidelines for BC

QUESTIONS ?




Thank
You